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A Process Algebra With Qualitative Calculus for Modeling Spatio-Temporal Behaviour

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ABSTRACT Distributive real-time autonomous systems like cyber-physical systems (CPS) and biology systems have become a hot topic nowadays. The high-level design of these systems shares a common characteristic that the behavior of agents is always space-related: i.e., agents are involved in an environment where the space relations between agents count. Agents are always sensitive to the discrete change of space relations among them at explicit time. We call such behavior the spatio-temporal behavior, which has become a critical research point recently. In this paper, we propose a process algebra called Communicating Sequential Process with Qualitative calculus (QCSP) as a formal language for modeling the spatio-temporal behavior. The key feature of our algebra is that it can reason about the space relations between agents by a finite set of binary relations that obey certain mathematical conditions — a qualitative calculus. In QCSP, each agent is confined in a space region. The space region (with an agent in it) can move as a whole and interact with other space regions. The topological structure of space regions forms the outer environment of agents, and the agents react timely according to their environment from time to time. We built our calculi by conservatively extending Timed CSP with qualitative calculus embedded in it, and illustrate how QCSP can be used to model the spatio-temporal behavior by analyzing three examples. We propose a special process algebra as a formal model for behavior that is sensitive to qualitative changing space and time. The key feature of our algebra is that it can reason about the relative location between agents by using qualitative calculus. Each agent is confined in a space region. The space region can move as a whole along with the agent confined in it and interact with other space regions. The behavior of agent depends on the interaction of space regions at an explicit time, shown by that certain actions can be triggered (or allowed) under certain space region relations and time condition. Such spatio-temporal behavior is a critical research point in nowadays distributive real-time autonomous systems like CPS or intelligent transportation systems (ITS). We propose our calculi by conservatively extending Timed CSP with qualitative calculus embedded in it to capture space region relations among agents.

INDEX TERMS Spatio-temporal behavior, process algebra, qualitative calculus, system modeling.

I. INTRODUCTION

Spatio-temporal behavior refers to computations that are highly sensitive to qualitative evolving space and time. They can be found in many distributive real-time autonomous systems such as CPS [1] (Cyber Physical Systems), Biology Systems [2], etc. The notion of ‘Space’ (or called ‘Space region’ or ‘region’) in these systems denotes physical or conceptual spaces such as points, locations, positions, places and so on. The ‘environment’ of agents, which represents

a type of relation between spaces, discretely evolves from time to time, while the agent is sensitive to and can react timely according to that evolvement. Consider the following 3 examples in the domain of CPS:

- 1) In a wireless sensor network [3] (Section VI-A), a temperature sensor can sense the temperature in the physical world. Here the ‘space’ is the temperature value (as a point on the real number line) and the ‘environment’ denotes the relations between temperature values (e.g., the temperature is more than 25 degree). The temperature keeps changing discretely following some patterns (e.g. it rises periodically to over 25 degree

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in 11 hours then falls to lower than 5 degree in the rest 13 hours of a day), while the sensor reacts timely according to which interval the temperature lies in.

- 2) In a mobile ad hoc network [4] (Section VI-B, Fig. 5), each node has a transmission range, within which its broadcasting messages can be received by other nodes. Here the ‘space’ denotes the transmission range, while the ‘environment’ means the topological structure of transmission ranges of the nodes, which keeps changing discretely as each node can move freely from one location to another. Nodes communicate with each other accordingly based on the topological structure of their transmission ranges at current time.
- 3) In an intelligent railroad crossing control system [5] (Section VI-C, Fig. 7), the ‘space’ means the location of the train and different areas on the track, while the ‘environment’ denotes the relative relation between the location of the train and different areas. The train keeps running on the track, it arrives certain areas where it interacts with the gate in order to pass the crossing safely.

In distributive real-time autonomous systems, the spatio-temporal behavior is addressed by hiding the unpredictable and unnecessary explicit variations of the agents’ environment. Engineers prefer to model the discrete ‘changing pattern’ of the environment rather than the explicit ‘motion locus’ when the explicit variation of the environment is hard or even impossible to predict in the system, or is irrelevant to the system behavior at some abstract design level. Such an ‘abstraction’ on the system’s behavior is important for the design of distributive real-time autonomous systems (like CPS and Biology Systems) at a high-level, and especially for modeling context-aware systems [6] where environment is open and the behavior of agents in the environment is hard to predict in advance. In the examples mentioned above, the temperature (in Example 1) is always changing unpredictably and impossible to be captured in reality. So we ignore the continuous change of temperature and only focus on the interval the temperature lies in from time to time. The motion locus of mobile nodes in an ad hoc network (of Example 2) is not necessary to analyze the capability of communications between nodes. Here only the topological structure between the transmission ranges of nodes matters. The same for the movement of a train running on a track (in Example 3). The explicit continuous movement of a train does not matter for the interactions between the train and the gate on specific locations.

In mathematics, the topology of spaces (i.e., the environment of agents) is often expressed with a finite set of relations, so that the environment of agents can be captured in a finite language [7]. For example, in Example 1 discussed above, the relation between the temperature and the degree value 25 and 11 can be captured by the relation ‘less than’ **LT** and ‘less than equal’ **LE** (see section VI-A for more details). The topological structure of transmission ranges in Example 2 needs to be described with a more complex set of

relations, like ‘overlap’, ‘communicate with’, ‘distant from’ and so on (see section VI-B for more details). These sets of relations equipped with certain mathematical structures are called ‘qualitative calculus’ [8], which has been used in expressing qualitative information in applications such as robot navigation [9], geographic information systems [10], etc. Qualitative calculus not only provides a finite expression for the topological structure of spaces, but also provides a formal way of reasoning it.

In this paper, we apply the qualitative calculus in building high-level formal model for spatio-temporal behavior in distributive real-time autonomous systems. We study the mobility of space regions related to each other in qualitative measures. Agent is confined in a space region, as a moving region with computations it carries. Agent can decide the movement of its own region and conversely the movement of regions can timely trigger the behavior of agent. Each agent owns a local view of region relations, which shows that how its region relates to other regions. We summarize the characteristics of spatio-temporal behavior as follows:

- 1) The environment of agent evolves dynamically from one state to another. The evolvement is reflected by the mobility of space regions. Moving region can be referred to many objects in different applications, as we have already seen from the above examples. But the referred objects must evolve (or must be abstracted to evolve) discretely in time, rather than continuously. For example, temperature that changes continuously from time to time cannot be taken as a space region.
- 2) Each agent is confined in one (and only one) space region. Agent is considered as a computation unit restricted in a bounded space.
- 3) Instead of declaring that an agent can move, we allow space region to move as a whole, carrying its computation—agent along with it. If an agent does not have an intuitive confined region, we just simply assume itself as a region. Agent can have a control of its region’s movement, but not of others’. The movement of regions is reflected by the changing structure of relations between regions.
- 4) A space region can contain more than one agent in itself. If a region does not contain any agents, it cannot move.
- 5) Each agent has its own environment, distinguished from other agents’. It is from a natural point that each agent, as an independent and distribute computation unit, cannot observe other region relations that have nothing to do with it.

Based on these characteristics, in this paper, we mainly propose a process algebra called QCSP (CSP with Qualitative calculus) to model spatio-temporal behavior. We extend Timed CSP [11] on its syntax by importing space regions and their relations as primitives, and adopt qualitative calculus [8] as a well-defined, complete relational model to reason about space relations in our calculi. With qualitative calculus one can directly reason about the movement and the evolving

TABLE 1. Comparisons between different space-related process algebras. Qualitative relation can actually include both ambient relation and communication relation. Qualitative relation can capture some metric relations. Their relations are: qualitative relation $\supset$ ambient relation $\supset$ communication relation, qualitative relation $\cap$ metric relation $\neq \emptyset$.

Process Algebra	Space		Mobility		Space Region Relation	Application Domain
	Real Space	Space Region	implicitly	explicitly		
CAM	×	✓	×	×	×	general
ACP _{ρσ}	✓	×	×	✓	metric space $\mathbb{R}^d$	general
π -calculus	×	×	✓	×	communication relation	general
AC	×	✓	×	✓	ambient relation	general
STeC	✓	×	×	✓	×	CPS
Space π -calculus	✓	×	×	✓	metric space $\mathbb{R}^d$	Biology System
ω -calculus	×	✓	×	✓	communication relation	Ad Hoc Network
CMC	×	✓	×	✓	ambient relation	Ubiquitous Computing
CaIT	×	✓	×	×	simple distance	Internet of Things
AbC	×	×	✓	×	predicate	Collective Adaptive Systems
CCA/typed CCA	×	✓	×	✓	ambient relation	Context-aware Systems
QCSP	✓	✓	×	✓	qualitative relation	general

environment of agents in QCSP. The rest of this paper is organized as follows:

- Section II introduces the related work of the proposed calculi. We compare previous works with our work and analyze their advances and limitations.
- In Section III, we introduce some pre-requisites about qualitative calculus. We introduce several examples of qualitative calculus that will be used in examples in Section VI.
- Section IV introduces the transition of space region relations in order to express the mobility of agents in our calculus. We define the dynamic mapping as a primitive to capture the dynamic evolvement of space regions.
- From Section V, we give the syntax and operational semantics of our calculus based on labelled transition systems, which forms the main contribution of this paper.
- In Section VI we analyze the applicability of QCSP by showing how QCSP model the 3 examples above.
- Section VII gives the conclusion and future work.

II. RELATED WORK

The process algebra for modeling space-related behavior has been fully investigated since 1990's. Most of the languages were proposed as a natural extension from the earliest classical process algebras: CSP [12] (Communicating Sequential Process), CCS [13] (Calculus of Communication System) and ACP [14] (Algebra of Communicating Processes). They have different ways of representing the notion of 'space', different abilities of expressing the notion of 'mobility', different understandings about 'space relations' and different domain applications. In the rest of this section we analyze the most famous general space-related process algebras and some recent specific space-related process algebras in terms of these 4 aspects (see Table 1 as an overview), and make a comparison between them and our proposed process algebra QCSP.

In Chemical Abstract Machine [15] (CAM), the author proposed a computation model based on 'chemical reactions' instead of traditional views where program is thought as a

sequence of executing procedures. In this model the notion of 'membranes' was imported as a 'confined region' of an agent (or say 'process'¹). Nevertheless, the membrane there is only a mechanism to make sure that the 'hiding operator' works normally and has a nature meaning as in traditional process algebras like CCS. It indicates no meanings about the agents' space and mobility. We adopt the 'chemical view' there in dealing with our calculus, where we take each process as confined in its space region.

Real space process algebra ACP_{ρσ} [16] based on ACP introduced locations with real values as an attribute of atomic actions. It indicates the actions that happen at an explicit location and time. In ACP_{ρσ}, the space denotes an explicit location in the Euclidean space $\mathbb{R}^d$ (i.e., a real value in domain $\mathbb{R}^d$), the space relation is the metric relation in Euclidean space $\mathbb{R}^d$. For example, in Example 2, node a is at location $l_1 = (0, 0, 0)$, node b is at location $l_2 = (0, 1, 0)$ in space $\mathbb{R}^3$, node a can communicate with node b when their distance $d(l_1, l_2) < 1$. Compared with ACP_{ρσ}, the space in QCSP denotes a more abstract notion, not limited to just physical points. QCSP can also represent the location and space relation in real space, but in a 'qualitative way'. e.g., we can use a relation $l_1 \text{ CMI } l_2$ (see Definition 6.1 below) to express $d(l_1, l_2) < 1$. However, the ability of QCSP to express metric relation in $\mathbb{R}^d$ is quite limited.

π -calculus [17] is a model to capture agents with moving behavior. The mobility of agents is implicitly expressed as the change of communication relation between agents. It extends CCS by simply removing all distinctions between channels and messages. Each agent can both send messages or channels to other agents. The change of communication channels from one agent to another indicates the mobility of agents. e.g., in Example 2, node b can send a 'channel' to node c so that c can communicate with b . However, the notion of space in π -calculus is implied by the change of

¹In some process algebras, 'process' just means the 'agent' here in our paper, i.e., an independent computation node in a network. We use 'agent' instead to denote the process in these process algebras. In our paper, 'process' means a local computation in an agent.

communication relation between agents. Since π -calculus itself does not specify space relations, there is no apparent restrictions to limit the evolvement of space under specific situations. Considering Example 2, assume node c is entering the transmission range of node a , while leaving the transmission range of node b . Using π -calculus we can express the movement of c by sending a channel from the node c to node a . But no mechanism can guarantee that node c can no longer communicate with node b since node b still holds node c 's channel. Moreover, the mobility in π -calculus is implicit, i.e., by only showing the change of communication relation one cannot tell which node (node c or node a) is moved. Compared with π -calculus, our QCSP explicitly defines the notion of space region and mobility. Each agent is confined in a space region and it moves as its region moves. The topological structure of space regions is explicit and captured as a set of qualitative relations between regions. The qualitative relation in QCSP is also able to express the communication relation in π -calculus.

Contrary to π -calculus, Ambient Calculus [18] (AC) clearly introduces 'ambience' notion as a first class citizen to indicate 'space' and introduces the movement notion—capability to indicate the movement of agent from one ambience into another. AC keeps the distinguish of channels and messages in the original CCS, i.e., not allowing channels as a type of messages transmitted between agents. The notion of 'space region' in QCSP was largely inspired by the 'ambience' in AC. Compared with the notion of ambience, space region is more general in terms of representing 'spaces'. In AC, ambiances can only be related to each other by ambient relations, in which ambience can only be 'bounded' or 'not bounded' by another ambience. e.g., in Example 2, one can only describe node a is 'in' the transmission range of b , or not. Because of this, the relation between ambiances forms a tree structure (any ambience has all ambiances which are bounded by it as its 'leave nodes'), which is suitable for modeling applications where the structure of spaces is a tree structure, e.g. the internet networks [18]. But this relations are not suitable for capturing more complex topological structures of spaces, like the structures shown in Example 2 above, where the range relations between nodes is not a tree structure (where the range of x_a and x_b could be intersected). Instead of the tree-structure-like ambience relations we adopt a more general structure—qualitative relation networks to express a more explicit relations between spaces. The space relations based on qualitative relations can reach a wider range of applications.

Other calculus proposed in recent years are largely based on π -calculus [17] and AC [18]. They are mainly for modeling applications in more specific domains. Their notions of space are hence more specific and closer to the explicit scenario, where different communication mechanisms and dynamics of spaces were defined and analyzed. Compared with them, QCSP is a general process algebra, whose space denotes a more general notion. In QCSP we focus on the finite expression of the topological structure of physical or

conceptional spaces, and how the structure affects the behavior of agents.

The location-time triggered language STeC [5], [19] is a process algebra for modeling CPS systems. It is based on time interval semantics. During a time interval, a process can move from one location to another. Their locations are based on 'pure names' such as the place name. No relations between locations are indicated.

Space π -calculus [20] is a model for Biology Systems. In space π -calculus space is modeled as the vector space in $\mathbb{R}^d$, the moving pattern of process is directly given by a movement function which is non-deterministic with parameters and varies in different abstraction level and application area.

ω -calculus [21] is a model for mobile ad hoc network systems. It is extended from π -calculus with a restriction of a set of groups on each process. Nodes in the same group are in each other's transmission range and thus can communicate with each other. The structure of groups evolves indicating the movement of nodes, which is similar to our approach.

CMC (Calculus of Mobility and Communication) proposed in [22] is a process calculi for ubiquitous computing. It is mainly based on AC [18], but is extended to include global form of communication, 'push and pull capabilities' and context awareness primitives in order to model special features of systems in ubiquitous computing.

The Calculus of the Internet of Things (CaIT) [23] is a process algebra for modeling the Internet of Things [24]. Its syntax is mainly built on CCS [13] with discrete time. Space in CaIT denotes an abstract physical location expressed by an integer number. The distance between locations is measured by an integer number. The communication capability between agents can be affected by the distance. As in QCSP, each agent is confined in a location. However, CaIT cannot specify the movement of agents, agents can only 'randomly' move from one location to another.

[25] proposed a calculus called AbC (Attributed-based Communication) and its behavioral theory for Collective-Adaptive Systems [26]. In AbC, the space is defined as the notion of 'attribute environment' — a set of attributes based on which the interaction between agents is dynamically established. The values of attributes can be dynamically modified, indicating the change of the environment. AbC used predicates to describe its attribute environment, while in QCSP we use qualitative calculus to describe the environment.

The Calculus of Context-aware Ambient (CCA) [27] was proposed in 2011 for modeling and verification of mobile systems that are context-aware. Its typed version [28] was proposed later to gain a further control of the communication between ambients. CCA inherited the notion of ambient as space from AC [18], and enriched AC with the 'context-aware capability' so that each agent can 'be aware of' the context (i.e., the environment) accordingly. There introduced 'context expression' to capture the relationship between ambients, which is similar to the introduction of qualitative calculus to express the relationship between space regions in our QCSP.

Compared with QCSP, the relation between ambients in CCA is simpler and are not able to deal with ambients with complex topological structures, just like the Example 2, 3 discussed above.

Except from space-related process algebras, there are also other types of formalisms for modeling and analyzing space-related behavior for real-time autonomous systems, such as I/O automata [29], Bigraphs [30] and Petri-Nets [31], [32]. Compared with process algebras, some models not only are used to conduct the high-level modeling, but also for modeling and analyzing detailed behavior of large-scale systems, e.g. [33].

III. SPACE REGIONS AND RELATIONS

The topological spatial (or temporal) relations have been fully investigated since 1980's. The earliest example is the work of using interval logics for the representation of time by Allen [34]. Since then many qualitative calculus [8] have been developed, such as point calculus [35], region connection calculus (RCC5/8) [36], etc. They all use dyadic relations as the basic measure to describe the topological relations on space or time, based on which a logic reasoning system can be built. It has been widely applied in many areas in artificial intelligence such as robot navigation, geographic information systems, etc.

A. QUALITATIVE CALCULUS

Let U be a non-empty universal set, consider a *partition scheme* of $U \times U$ into a finite set of non-empty and disjoint binary relations $(r_i)_{i \in I}$:

$$U \times U = \bigcup_{i \in I} r_i$$

where I is a finite index set. Every element $(x, y) \in U \times U$ must and only belongs to one of r_i . $(r_i)_{i \in I}$ is called *basic relations*. $\mathcal{B}$ is often used to denote the set of basic relations.

It is assumed that the identity relation r_0 belongs to $(r_i)_{i \in I}$:

$$r_0 = \{(x, y) \mid (x, y) \in U \times U, x = y\}.$$

Sometimes $(x, y) \in R$ is abbreviated as xRy , R is a relation. The inverse of a relation R is defined as R^{-1} such that:

$$R^{-1} = \{(x, y) \mid (y, x) \in R\}.$$

The composition of two relations R_i, R_j is defined as:

$$R_i \circ R_j = \{(x, y) \mid \exists v \in U. (xR_iv \wedge vR_jy)\}.$$

A boolean algebra $\mathcal{A}$ is of all subsets of basic relations $(r_i)_{i \in I}$, with the operators between sets: union ($\cup$), intersection ($\cap$) and complementation ($-$). Each element of $\mathcal{A}$ is either a basic relation r_i , or a relation of an union of several basic relations. The top element of $\mathcal{A}$ is denoted as $\mathbf{1}$ (the total set), called the 'universal relation', while its bottom element is denoted as $\mathbf{0}$ (the empty set), called the 'empty relation'.

If a boolean algebra $\mathcal{A}$ is closed under inverse and composition, it is then called *relational algebra*, an example of which is point algebra (Section III-B.1). A relation R is

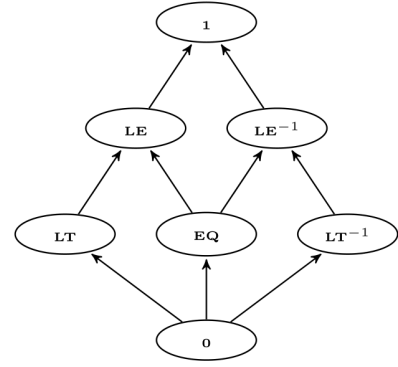


FIGURE 1. The lattice structure of $\mathcal{A}_{PA}$.

closed under an unary operator F on relations if $F(R) \in \mathcal{A}$. A binary operator can be seen as an unary operator by fixing one parameters. For example, the composition can be seen as an unary operator in the form of $(R' \circ)(R) = R' \circ R$ or $(\circ R')(R) = R \circ R'$ for all $R' \in \mathcal{A}$.

A *qualitative calculus* is defined as a boolean algebra $\mathcal{A}$ which is closed under inverse. In qualitative calculus, there are often some assumptions, for instance, as we will see in the interval algebra of Example 3 (Section VI-C, assumption p_3), it is assumed that all space regions in the railroad crossing are connected to each other. Thus in this paper, we understand qualitative calculus as a pair $\langle \mathcal{A}, \mathcal{A}_s \rangle$ where $\mathcal{A}_s$ is a set of assumptions.

B. EXAMPLES OF QUALITATIVE CALCULUS

In this section, we give two examples of qualitative calculus that will be used in section VI.

1) POINT ALGEBRA

A simple example of qualitative calculus is point algebra [35], denoted by $\langle \mathcal{A}_{PA}, \mathcal{A}_{sPA} \rangle$. Given a region set U , relation $x\mathbf{LT}y$ indicates the traditional 'less than' operator in $\mathbb{R}$, denoted by $<$, read as 'x is less than y'.

Other relations can be naturally defined:

$$x\mathbf{LE}y \triangleq \neg(y\mathbf{LT}x);$$

$$x\mathbf{EQ}y \triangleq x\mathbf{LE}y \wedge y\mathbf{LE}x.$$

$x\mathbf{LE}y$ means x is less than or equal to y (corresponding to $x \leq y$ in $\mathbb{R}$). $x\mathbf{EQ}y$ means x is equal to y (corresponding to $x = y$).

The basic relation set is $\mathcal{B}_{PA} = \{\mathbf{LT}, \mathbf{EQ}, \mathbf{LT}^{-1}\}$. $\mathbf{LT}^{-1}$ is actually the inverse of $\mathbf{LT}$, corresponding to $>$. Similar for $\mathbf{LE}^{-1}$, which corresponds to $\geq$. A boolean algebra can be acquired by $\mathcal{B}_{PA}$, denoted as $\mathcal{A}_{PA}$. Fig. 1 shows a lattice structure of $\mathcal{A}_{PA}$ ordered by set inclusion $\subset$. Where non-basic relations are given as: $\mathbf{LE} = \mathbf{LT} \cup \mathbf{EQ}$, $\mathbf{LE}^{-1} = \mathbf{LT}^{-1} \cup \mathbf{EQ}$. $\mathbf{1} = \mathbf{LT} \cup \mathbf{EQ} \cup \mathbf{LT}^{-1}$ is the total relation.

It is easily checked that $\mathcal{A}_{PA}$ is a relational algebra since it is closed under inverse and composition. For instance,

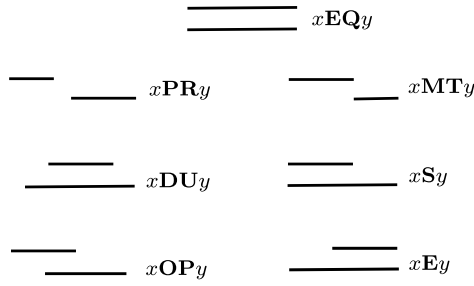


FIGURE 2. Relation of intervals for each base relation.

$LT \circ LT = LT$, $LE \circ LE = LE$, $LT \circ EQ = LE$, $LE \circ EQ = LE$, $EQ \circ EQ = EQ$, etc.

No assumptions in point algebra so $As_{PA} = \emptyset$.

2) INTERVAL ALGEBRA

Interval algebra [34] is built on a linear order set S . The interval as a primitive object is constructed as ordered pairs of distinct ‘points’:

$$\langle a, b \rangle = \{c \mid a, b, c \in S, a \leq c \leq b\}.$$

U is a collection of intervals, ranged over by x, y :

$$U = \{\langle a, b \rangle \mid a, b \in S, a < b\}.$$

The basic relations between intervals are defined based on the partial order relations between ‘endpoints’:

$$\langle a, b \rangle EQ \langle a', b' \rangle \triangleq a = a' \wedge b = b';$$

$$\langle a, b \rangle PR \langle a', b' \rangle \triangleq a < b < a' < b';$$

$$\langle a, b \rangle DU \langle a', b' \rangle \triangleq a' < a < b < b';$$

$$\langle a, b \rangle O \langle a', b' \rangle \triangleq a < a' < b < b';$$

$$\langle a, b \rangle MT \langle a', b' \rangle \triangleq a < b = a' < b';$$

$$\langle a, b \rangle S \langle a', b' \rangle \triangleq a = a' < b < b';$$

$$\langle a, b \rangle E \langle a', b' \rangle \triangleq a < a' < b = b'.$$

The explanation of each relation is illustrated in Fig 2: $xEQy$, two intervals x and y are equal. $xPRy$, x is ahead of y without interleaving. $xDUy$, x is partially in y . xOy , x, y has an intersecting part in the middle. $xMTy$, x meets y at a single point in the middle. xSy , x starts at the beginning of y . xEy , x ends at the ending of y .

$\mathcal{B}_{IA} = \{EQ, PR, DU, O, MT, S, E, PR^{-1}, DU^{-1}, O^{-1}, MT^{-1}, S^{-1}, E^{-1}\}$ forms the basic relations in interval algebra $\langle \mathcal{A}_{IA}, As_{IA} \rangle$. There are total $2^{13} = 8192$ relations in $\mathcal{A}_{IA}$, thus instead of denoting them by names we use a set of basic relations to represent an union relation. For example, $x\{O, S, E\}y = xOy \vee xSy \vee xEy$.

Obviously, $\mathcal{A}_{IA}$ is closed under inverse. It is also closed under composition, for example, $O \circ O = \{PR, MT, O\}$, $MT \circ MT = PR$, $S \circ S = S$, etc. So it is a relational algebra.

The endpoints of an interval $x = \langle a, b \rangle$ is usually denoted as $x^+ = a$, $x^- = b$.

C. CONSTRAINTS NETWORK

A constraints network [8] $\nu : U' \times \mathcal{A} \rightarrow U'$ is a set of dyadic relations between elements of a subset U' of U . $\nu(x)(R) = y$ iff xRy holds for any $x, y \in U'$, $R \in \mathcal{A}$. A set of constraints networks is denoted by **Net**.

We use $Reg(\nu)$ to compute the set of elements U' for a network ν .

IV. TRANSITION OF SPACE REGION RELATIONS AND DYNAMIC MAPPING

Our calculi allows the mobility of space regions from one location to another. The movement of regions is implicitly reflected by the transition of region relations. Such transition may involve several changes on relations. For example, in interval algebra, region x moves into region y and z together at the same time. The process concerns two changes in constraints network—two new relations $xDUy$ and $xDUz$ are added into the network. To reduce complexity, we consider the relation changes as ‘one-by-one’ changes. We call it *one-step movement*. The movement of region x is divided into two steps—firstly moves into region y , then moves into region z , which corresponds to two independent changes in constraints network: firstly $xDUy$ is added, then $xDUz$ is added. The order of movements is non-essential since they happen at the same time.

A new relation is added into the constraints network when an one-step movement happens. However the new relation in the network may cause contradictions. For example, consider a network $\nu = \{xPRy, zMTy, xDUz\}$. Once $xDUy$ is added, the network becomes $\nu' = \{xDUy, xPRy, zMTy, xDUz\}$. It leads two contradictions: $xDUy, xPRy \vdash false$ and $xDUy, xDUz, zMTy \vdash false$. The first one is easy to see, since DU and PR are disjoint. For the second one, on one hand we have $x^- < z^- < y^+$ because of $xDUz$ and $zMTy$, on the other hand we have $y^+ < x^+ < x^-$ because of $xDUy$.

Contradictions not only arise from constraints network itself, but from the assumptions of qualitative calculus we give in advance. For example, if we assume that $xPRy$ always holds for some region x and y in interval algebra, then $\nu, xDUy \vdash false$ for any ν .

The deduction $\vdash$ depends on the qualitative calculus we give from the context.

Definition 4.1 (Qualitative Calculus Proof System): For a qualitative calculus $\langle \mathcal{A}, As \rangle$, a proof system can be formed: All axioms of this proof system consist of assumptions As and all tautologies in the classical first-order predicate logic system. The inference rules of this proof system consist of the conventional *Gen* and *MP* rules. The inference is denoted by $\vdash_{\mathcal{A}, As}$.

The new network acquired by adding a new relation into an original network should not contain those relations that cause contradictions under the proof system of qualitative calculus. A network without any deduced contradictions is called *consistent network*, as defined below:

Definition 4.2 (Consistent Network): For a network ν and a qualitative calculus $\langle \mathcal{A}, As \rangle$, if $\nu \not\vdash_{\mathcal{A}, As} false$, i.e., any

contradictions cannot be proved under the proof system of $\langle \mathcal{A}, As \rangle$, then call v a consistent network under $\langle \mathcal{A}, As \rangle$, denoted as $v^c: \langle \mathcal{A}, As \rangle$.

A set of consistent networks is denoted by $\mathbf{Net}^c: \langle \mathcal{A}, As \rangle$.

We sometimes omit the qualitative calculus $\langle \mathcal{A}, As \rangle$ and simply write v^c and $\mathbf{Net}^c$ if $\langle \mathcal{A}, As \rangle$ is arbitrary or is clear from the context.

When causing contradictions after a new adding relation, it is a problem which relation should be removed from the original network in order to keep the consistency. Two basic principles should be followed: Firstly, naturally the movement of a region can not violate the assumptions of qualitative calculus first; secondly, as a region moves so that a new relation is formed, other regions should be static, thus the relations not concerning the moving region should not be removed. The moving region should be identified when we add a new relation into a network. We use a pair $\langle x, x_1 R x_2 \rangle$ to represent the new addition of relation. We use a notation $x \sim_R y$ to denote either $x R y$ or $y R x$ for any relation R .

Following the two principles above, after we add a new relation, e.g. $\langle x, x \sim_R y \rangle$, into a network, we need to check if $x \sim_R y$ violates the assumptions. If it does, we simply remove $x \sim_R y$. Otherwise we keep it and compute the *maximum consistent network* in the sense that keeping all relations that do not concern x unchanged in the original network.

However it comes ahead that whether such a maximum consistent network exists in any network if we keep $x \sim_R y$ and all relations that do not concern x unchanged. To answer this question we divide it into two sub questions: 1) whether a network consisting of $x \sim_R y$ and all relations $x_1 R' x_2$ not concerning x ($x_1 \neq x \neq x_2$) in the original network is consistent; 2) based on this network, whether we can get an unique largest consistent network by adding the rest relations $y_1 R'' y_2$ in the original network concerning x ($y_1 = x \vee y_2 = x$) into this network. The two sub questions are positive, stated and analyzed formally below.

Property 4.1: Given a qualitative calculus $\langle \mathcal{A}, As \rangle$ and a network v with a new addition $\langle x, x \sim_R y \rangle$ added into it (i.e. $x \sim_R y \in v$). If $v - \{x \sim_R y\}$ is consistent under $\langle \mathcal{A}, As \rangle$, then $v - (v' - \{x \sim_R y\})$ either violates As or is consistent under $\langle \mathcal{A}, As \rangle$, where $v' = \{y_1 R y_2 \mid y_1 R y_2 \in v, y_1 = x \vee y_2 = x\}$.

Property 4.2: Given a qualitative calculus $\langle \mathcal{A}, As \rangle$ and a network v with a new addition $\langle x, x \sim_R y \rangle$ added (i.e. $x \sim_R y \in v$). If $v - \{x \sim_R y\}$ is consistent under $\langle \mathcal{A}, As \rangle$, then based on $v - (v' - \{x \sim_R y\})$ there exists a maximum consistent network v'' of v under $\langle \mathcal{A}, As \rangle$ such that:

$$v'' = \begin{cases} v - \{x \sim_R y\}, & \text{if } v \vdash_{\mathcal{A}, As} \text{false} \\ (v - (v' - \{x \sim_R y\})) \cup \bigcup v, & \text{otherwise,} \end{cases}$$

where $v \subseteq v' - \{x \sim_R y\}$ and $(v - (v' - \{x \sim_R y\})) \cup v$ is consistent under $\langle \mathcal{A}, As \rangle$. The 'maximum network' is defined in the sense that for any consistent network v^* of v , $v^* \subseteq v''$.

We denote such a maximum network of v by $\mathbf{cnmax}^{\langle \mathcal{A}, As \rangle}(v, \langle x, x \sim_R y \rangle)$, which is acquired by keeping $x \sim_R y$ and all relations not concerning x in v .

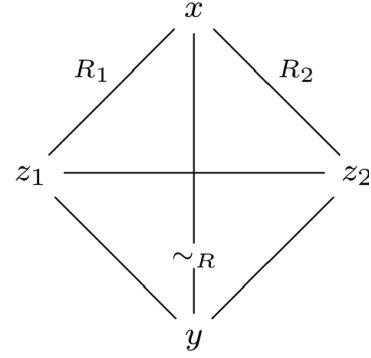


FIGURE 3. Triangle relations between x, z_1, z_2, y .

We write $\mathbf{cnmax}(v, \langle x, x \sim_R y \rangle)$ if the qualitative calculus is clear from the context and can be neglected.

The correctness of Property 4.1 is direct. Intuitively, if there is only one relation $x \sim_R y$ in a network that concerns region x , for other relations not concerning x , e.g. $y R' z$, no matter what R' is, $y R' z$ is impossible to contradict $x \sim_R y$.

As for Property 4.2, we only need to prove the existence of the maximum network since $v - (v' - \{x \sim_R y\})$ is assumed to be consistent. The key point is to prove that for any $v_1, v_2 \in v' - \{x \sim_R y\}$, if both $v_m \cup \{v_1\}$ and $v_m \cup \{v_2\}$ are consistent, then $v_m \cup \{v_1, v_2\}$ is also consistent, where $v_m = (v - (v' - \{x \sim_R y\})) \cup v'$ for some $v' \subseteq \bigcup v$. It is the fact that adding $x \sim_R y$ does not conflict the coincidence between v_1 and v_2 . To simply understand this, we can consider Fig. 3 below. Suppose $v_1 = x R_1 z_1$, $v_2 = x R_2 z_2$. Knowing that all contradictions must happen in a 'triangle relation' in network, it is easy to see that all triangles in the network that concern x, z_1, z_2 and y are consistent according to assumptions: triangle $x - z_1 - y$ and $x - z_2 - y$ are consistent since $v_m \cup \{v_1\}$ and $v_m \cup \{v_2\}$ are consistent, triangle $x - z_1 - z_2$ and $z_1 - z_2 - y$ are consistent since $v - \{x \sim_R y\}$ is consistent.

After showing the existence and uniqueness of the maximum consistent network after a new adding relation, we define a basic 'operator' indicating this change of networks in order to express 'the mobility of space region'. The dynamic mapping maps a consistent network to the maximum consistent network after an addition of a new relation into it.

Definition 4.3 (Dynamic Mapping): A dynamic mapping $[x, x \sim_R y]^{\langle \mathcal{A}, As \rangle} : \mathbf{Net}^c: \langle \mathcal{A}, As \rangle \rightarrow \mathbf{Net}^c: \langle \mathcal{A}, As \rangle$ is a function between two sets of consistent networks under $\langle \mathcal{A}, As \rangle$. It maps every $X \in \mathbf{Net}^c: \langle \mathcal{A}, As \rangle$ to its maximum consistent network under the sense of Property 4.2:

$$[x, x \sim_R y]^{\langle \mathcal{A}, As \rangle}(v) = \mathbf{cnmax}^{\langle \mathcal{A}, As \rangle}(v \cup \{x \sim_R y\}, \langle x, x \sim_R y \rangle).$$

A simple example can be shown as above where $v = \{x \mathbf{PR} y, z \mathbf{MT} y, x \mathbf{DU} z\}$. We add a new relation $\langle x, x \mathbf{DU} y \rangle$ to the network. Following Property 4.2, we obtain the maximum consistent network as:

$$\begin{aligned} [x, x \mathbf{DU} y](v) &= \mathbf{cnmax}(v \cup \{x \mathbf{DU} y\}, \langle x, x \mathbf{DU} y \rangle) \\ &= \{x \mathbf{DU} y, z \mathbf{MT} y\}. \end{aligned}$$

In our calculus (Section V), each agent owns a constraints network indicating a local graph of the total relations of all regions. It concerns only the region relations that relate its own region. Thus a constraint network can be split into several sub networks, each one reflects only a ‘local view’ of one region. We introduce the factorization of a network that will be used in our calculus.

Definition 4.4 (Factorization of Network by Region): Given a network ν under a qualitative calculus $\langle \mathcal{A}, \mathcal{A}_s \rangle$, we define a function $Fac : \mathbf{Net} \rightarrow \mathcal{P}(\mathbf{Net})$ as:

$$Fac(\nu) = (x[\nu])_{x \in Reg(\nu)},$$

where $\mathcal{P}(X)$ is the power set of a set X . $x[\nu] = \{x \sim_R y \mid R \in \mathcal{A}\}$ denotes the network ‘projected’ on region x .

V. INTRODUCTION OF QCSP

Our process algebra is largely based on the syntax of Timed CSP, from which many notions, symbols and operators are inherited. We omit the discussions about them in our paper, readers can refer to [11] [37] [38] for more details.

A. SYMBOLS

Atomic actions in QCSP are ranged over by a, b . The set of atomic actions is denoted by Σ . A special type of atomic actions is the dynamic mapping, as introduced in Section IV. They are named by the new additions of relation embraced by a pair of angle brackets, e.g., $\langle x, xDUy \rangle$. We range them over $\tilde{a}, \tilde{b}$. A special symbol Ξ is used to denote the set of all dynamic mappings in order to make a distinguish of the two types of atomic actions. α, β are used to denote any atomic actions in $\Sigma \cup \Xi$.

Processes in QCSP are ranged over by P, Q .

Communication is defined the same as in CSP [12]. $a.v$ means that a process inputs or outputs a value v through channel a . $a!v$ indicates a process inputs value v , while $a?v$ indicates a process outputs value v .

As indicated in section I, we abstract the notion of mobility of processes by introducing *space regions*. The relations between regions are captured under a type of qualitative calculus as introduced in Section III. A non-empty set of universal regions is assumed by U , x, y are used to range over regions. Each process P can be bounded with one specific region and its relations to other regions as a constraints network. e.g. $P : x[\nu]$ means that process P is in region x , its relations to other regions is expressed as the network $x[\nu]$.

A process is also an *agent* if it is considered as an independent and distributive unit, denoted by M, N . Not like process, an agent can only be composed by parallel operator $|$ indicating that agents execute in parallel. e.g. $|^A(M, N), |^A(M_1, M_2, M_3)$.

To capture the spatio-temporal behavior, we bind each atomic action with a *space condition* and a *time condition*. A space condition has a form of ‘ Ry ’ or ‘ yR ’, where $y \in U$, $R \in \mathcal{A}$. Time condition is just a time value $t \in \mathbb{R}^+$. For example, $a@(Ry, t)$ is an action bounded with a space

condition Ry and a time condition t . Agent $(a@(Ry, t) \rightarrow P) : x[\nu]$ means that action a can only occur at time t when region x (of the process) has a relation R with region y (i.e., xRy). An action can have more than one space conditions, linked by *and* operators, but only one time condition. e.g. $a@(R_1 y_1 \wedge y_2 R_2 \wedge R_3 y_3, t)$.

B. SYNTAX

We begin by first declaring some general assumptions about the systems we assume in our model:

1. *Instantaneous Events*: Each action occurs instantaneously at an exact time point with no time consuming. The communication between processes instantly happens at the same time the corresponding action occurs. We say ‘an action is ready to engage’ means that it is forced to happen once it is allowed by the environment. We allow an action wait for an arbitrary amounts of time for the allowance of environment.
2. *Global Clock*: A conceptual global clock is assumed to record the passage of time in total sense. All processes of system agree on this global clock to record their time elapsing.
3. *Continuous Time*: Time is modelled as the non-negative real numbers in $\mathbb{R}^+$, which means that there is no minimum granularity of time and there is no minimum interval between any two actions happened at different times.
4. *Realism*: We postulate that a system can only behave a finite set of actions in a finite time interval. It means that we only focus on the divergent timed traces (timed sequences with the limit of $+\infty$) in both operational and denotational semantics of our proposed calculus.
5. *Abstraction/Hiding*: When an action is hidden (or say abstracted), it becomes an internal action which environment do not observe. Once an internal action is ready to engage, it is forced to immediately happen since it does not need to wait for the allowance from the environment.

The assumption 1-3, 5 turn out to be the same assumptions for Timed CSP. We relist them here (for more details refer to [11] [38]). For assumption 4, not like in Timed CSP, we do not assume the minimal δ delay for every atomic action. Instead, we reach the same goal as in Timed CSP by only focusing on the divergent timed trace model (the same approach adopted in [39]). One can easily see the same meaning by different assumption comparing the one in [11].

We declare the syntax of QCSP in a BNF form in Table 2, where $\sim_R y$ denotes Ry or yR for any relation $R \in \mathcal{A}$, often written as $\sim_R y \in \{Ry, yR\}$. I is a finite set of indexes.

The syntax of QCSP is divided into two parts. Process ‘ P ’ is the basic component of a system. Agent ‘ M ’ is defined as a process restricted in a space region. Finally a system ‘ S ’ is defined as a sequence of agents running in parallel. The set of all processes and agents are denoted by $\mathcal{P}$ and $\mathcal{M}$ respectively.

E , $STOP$, $SKIP$ and $WAIT$ are *special processes*. E is an empty process, we assume $P=P < C > E=E < C >$

TABLE 2. Syntax of QCSP.

$P ::=$	$E \mid STOP \mid SKIP \mid WAIT \delta \mid$ $a \rightarrow P \mid a.v \rightarrow P \mid a@(\wedge_{i \in I} \sim_{R_i} y_i, t) \rightarrow P \mid$ $a@(\wedge_{i \in I} \sim_{R_i} y_i, t).v \rightarrow P \mid [x, x \sim_R y] \rightarrow P \mid$ $P <C> P \mid P \sqcap P \mid P \sqcup P \mid$ $P; P \mid$ $P \parallel_B P \mid$ $P \triangleleft (\sqcap_{i \in I} a_i \rightarrow P_i) \mid P \triangleleft P_1 <C> P_2 \mid$ $P \backslash A \mid$ $\mu X.P(X) \mid$ $f(P) \mid f^{-1}(P)$	(process) (special process) (prefix) (choice) (sequence) (parallel) (interruption) (abstraction) (recursion) (renaming)
$M ::=$	$P \mid P : x[v] \mid$ $\mid (M_1, M_2, \dots, M_n)$ A	(agent) (region) (ag parallel)

$P=E \sqcap P=P \sqcap E=P \sqcap E=E \sqcap P=P; E=E; P=E \triangleleft P$. A process that proceeds *STOP* does nothing, but waiting forever not allowing any actions. *SKIP* indicates a successful termination of a process. Process *WAIT* δ just allows the time passing δ time while no actions can be executed. *WAIT* ∞ behaves as a process that is forever waiting. For any $t \in \mathbb{R}$, there is $\infty - t = \infty$.

prefix $a \rightarrow P$ links an atomic action (in Σ) with a process, it is the sequentiality of a process that firstly proceeds a then proceeds the rest part of it. An atomic action a can be either bounded with a value v (in the form of $a.v$) or a space-time condition ($\wedge_{i \in I} \sim_{R_i} y_i, t$) (in the form of $a@(\wedge_{i \in I} \sim_{R_i} y_i, t)$) or both (in the form of $a@(\wedge_{i \in I} \sim_{R_i} y_i, t).v$). We also write $a@(\wedge_{i \in I} \sim_{R_i} y_i)$ if the time condition is *true* and $a@(t)$ if the space condition is *true*.

Prefix $[x, x \sim_R y] \rightarrow P$ is the form of dynamic mappings. As one agent moves, its region moves together. Each agent can only perform the movement within its own region, which is to say, e.g. $([z, x \sim_R y] \rightarrow P) : x[v]$ can be only performed if $x = z$. Thus sometimes under a region restriction of agent we can omit the region x as the form: $([Ry] \rightarrow P) : x[v]$ or $([yR] \rightarrow P) : x[v]$ depending on $x \sim_R y$.

$P <C> Q$ is a *conditional choice* where $C \in \{true, false\}$ is a boolean value. If $C = true$, P is selected, otherwise Q is selected. $P \sqcap Q$ expresses the *external choice*. From the view of observers, it waits for the environment to determine the choice of P or Q , if the environment can not determine (the case that the next action of P is the same as that of Q), it nondeterministically selects one of them. $P \sqcup Q$ is called *internal choice*. It either nondeterministically selects P or Q if the next actions of them are allowed, or selects none of them (even if their next actions are allowed). Operators $\sqcap$ and $\sqcup$ offer to describe the behavior of system on different abstraction levels by allowing internal actions ' τ ' (invisible for observers), which will be given in the operational semantics of QCSP.

The *sequence* operator $P; Q$ describes the sequentiality of two subprocess P and Q where Q can proceed only when P successfully terminates.

$P \parallel_B Q$ means that two processes P and Q execute in parallel, with synchronous communication if both P and Q offer a same next action. $A, B \subseteq \Sigma \cup \Xi$ are sets of

atomic actions. Process P, Q are permitted to proceed their own actions alone in $A - B, B - A$ respectively, and to communicate if they offer actions in $A \cap B$.

Interruption describes a jumping behavior from a continuous executing state of one process to another. $P \triangleleft (\sqcap_{i \in I} a_i \rightarrow P_i)$ means that process P is stopped and any action a_i proceeds if the environment offers a_i . While $P \triangleleft (\sqcap_{i \in I} P_i)$ means that P may be stopped and any process P_i proceeds (if it is allowed), but it may also be that P continuously proceeds without any interruption. $P \triangleleft (P_1 <C> P_2)$ means that if $C = true$, P is stopped and P_1 proceeds, otherwise P_2 proceeds after P is finished.

Abstraction 'hides' actions from the view of outside behaviors. If an action is an internal one from outside, the environment cannot participate in that action. Abstraction allows different levels of models of a system, it builds an abstract behavioral model of a system so that from the view of observers, the behavior of that system is different from the original one.

Recursion is the least fixed point (denoted by $\mu X.P(X)$) of the function $X = P(X)$. Recursion $\mu X.P$ repeatedly proceeds P for infinitely many times: $\mu X.P = P[\mu X.P/X]$ by replacing each X appeared in P with $\mu X.P$. For example, $P = \mu X.(a \rightarrow X) = a \rightarrow (\mu X.(a \rightarrow X)) = a \rightarrow a \rightarrow (\mu X.(a \rightarrow X)) = \dots$

Renaming f maps each atomic action in Σ to another one, f^{-1} is the inverse of f . The mapping on process and agent can be deductively defined based on the syntax structure. For example, let $f(a) = a', f(b) = b'$, then $f(a \rightarrow b \rightarrow STOP) = a' \rightarrow f(b \rightarrow STOP) = a' \rightarrow b' \rightarrow f(STOP) = a' \rightarrow b' \rightarrow STOP$.

Agent M is defined as a process which is restricted in a space region. $P : x[v]$ indicates that process P is confined in region x , with a constraint network $x[v]$ indicating the region relations between x and others. An agent can be a process not restricted by any region. $Loc(M)$ returns the network of agent M . e.g. $Loc(P : x[v]) = x[v]$.

The parallel for agents $\mid (M_1, M_2, \dots, M_n)$ is a n -tuple function that describes n agents A running in parallel. It is similar to the one defined for processes. The difference is that the execution of dynamic mapping action in one agent would lead the change of constraints networks of all agents, and the communication between agents turns out to be broadcast among these agents. This will be fully illustrated in the operational semantics of QCSP.

C. OPERATIONAL SEMANTICS

In the last section we give an intuitive meaning of each formula. In this section we formalize these meanings by giving the operational semantics of QCSP.

Like in Timed CSP, we introduce $\surd$ and τ to express the action of *SKIP* and the internal action respectively. We also define the action sets as follows:

$$\begin{aligned} \Sigma^\tau &= \Sigma \cup \{\tau\}; \Sigma^\surd = \Sigma \cup \{\surd\}; \Sigma^\tau, \surd = \Sigma \cup \{\tau\} \cup \{\surd\}; \\ \Xi^\tau &= \Xi \cup \{\tau\}. \end{aligned}$$

We use θ to denote actions in Ξ^τ , μ to denote actions in $\Sigma^\tau \cup \Xi$, e to denote actions in $\Sigma^\vee$, and ω to denote actions in $\Sigma^\tau, \vee \cup \Xi$.

As indicated in the global assumptions we listed above, once an action is ready to engage, it is forced to occur if it is accepted by environment. We introduce the initial engagement of a process defined as below. For the same operators appeared in Timed CSP, the rules are directly inherited from the definition of engagement in [39]:

Definition 5.1 (Initial Engagement of QCSP): The initial engagement for a process is a function $init(P)$ that maps each process P to a set of atomic actions in which P is ready to engage next. It is defined as follows based on the syntactic structure of QCSP:

- 1) $init(E) = init(STOP) = init(WAIT \ \delta) = init(\mu X.P(X)) = \emptyset$, where $\delta > 0$.
- 2) $init(SKIP) = init(WAIT \ 0) = \{\sqrt{\cdot}\}$.
- 3) $init(\alpha \rightarrow P) = \{\alpha\}$.
- 4) $init(P <C> Q) = P$ if $C = true$, $init(P <C> Q) = Q$ if $C = false$.
- 5) $init(P \sqcap Q) = init(P) \cup init(Q)$.
- 6) $init(P \sqcap Q) = \emptyset$.
- 7) $init(P; Q) = init(P) - \{\sqrt{\cdot}\}$.
- 8) $init(P \ A \ B \ Q) = (init(P) \cap (A - B)) \cup (init(Q) \cap (B - A)) \cup (init(Q) \cap init(P) \cap (A \cap B))$.
- 9) $init(P \triangleleft (\bigcap_{i \in I} a_i \rightarrow P_i)) = init(P) \cup \bigcup_{i \in I} \{a_i\}$.
- 10) $init(P \triangleleft (\bigcap_{i \in I} P_i)) = init(P)$.
- 11) $init(P \triangleleft (P_1 <C> P_2)) = init(P_1)$ if $C = true$, $init(P \triangleleft (P_1 <C> P_2)) = init(P)$ if $C = false$.
- 12) $init(P \setminus A) = init(P) - A$.
- 13) $init(f(P)) = f(init(P))$, $init(f^{-1}(P)) = f^{-1}(init(P))$.
- 14) $init(P : \langle x, v \rangle) = init(P)$.
- 15) $init(\bigcup_A (P_1 : \langle x_1, v_1 \rangle, \dots, P_n : \langle x_n, v_n \rangle)) = \bigcup_{i=1}^n (init(P_i) \cap ((\Sigma^\vee \cup \Xi) - A)) \cup (\bigcap_{i=1}^n init(P_i) \cap A)$.

Like the operational behavior of Timed CSP, once concerning a continuous time model, we need to consider two types of transitions in a transition system:

1. **Continuous Transition:** The evolution from one configuration to another by just allowing time to pass.
2. **Labelled transition:** The instantaneous jumping from one configuration to another by triggering an action at a particular time point.

The labelled transition system based on the two types of transitions are given in a natural way:

Definition 5.2 (Labelled Transition System of QCSP): A labelled transition system, denoted as $\mathcal{TS} = (\mathcal{M}, \rightarrow, \Rightarrow)$, takes agents as states, and takes function $\rightarrow: \mathcal{M} \times \Sigma^\tau, \vee \times \mathbb{R}^+ \rightarrow \mathcal{M}$ as labelled transitions, function $\Rightarrow: \mathcal{M} \times \mathbb{R}^+ \rightarrow \mathcal{M}$ as continuous transitions between states.

Given a finite set of agents $(M_i)_{i=1,2,\dots,n} \subseteq \mathcal{M}$ ($n \geq 1$), The reflective transitive closure of finite continuous transitions is defined as $M_1 \xRightarrow{\delta}^* M_n$ if

$$M_1 \xRightarrow{\delta_1} M_2 \xRightarrow{\delta_2} \dots \xRightarrow{\delta_{n-1}} M_n,$$

where $\delta = \sum_{k=1}^{n-1} \delta_k$. The reflective transitive closure of finite labelled transitions is defined as $M_1 \xrightarrow{\omega_1 \omega_2 \dots \omega_m}^* M_n$ if

$$M_1 \xRightarrow{\delta_0}^* M_{i_1} \xrightarrow{\omega_1} M_{i_1+1} \xRightarrow{\delta_1}^* M_{i_2} \xrightarrow{\omega_2} M_{i_2+1} \dots M_{i_{m-1}+1} \xRightarrow{\delta_{m-1}}^* M_{i_m} \xrightarrow{\omega_m} M_{i_m+1} \xRightarrow{\delta_m}^* M_n,$$

where $1 \geq i_1 < \dots < i_m < n$, $m \geq 1$.

We list below the operational semantics of QCSP for each formula. For all rules listed below, we assume that $\delta, \delta' \geq 0$ everywhere. We omit the region restrictions ($: x[v]$) for all agents whose transitions do not concern the change of the region relations.

$$1. \frac{}{STOP \xRightarrow{\delta} STOP} \text{ (Stop)}$$

$STOP$ can wait for any δ time and stays in $STOP$ forever.

$$2. \frac{}{SKIP \xRightarrow{\delta} SKIP} \text{ } (\delta - Skip)$$

$$\frac{}{SKIP \xRightarrow{\sqrt{\cdot}, t} E} \text{ (Skip)}$$

Rule $\delta - Skip$ means that $SKIP$ can wait for any δ time. Rule $Skip$ shows that at any time t , $SKIP$ can be triggered and produces $\sqrt{\cdot}$.

$$3. \frac{\delta' < \delta}{WAIT \ \delta \xRightarrow{\delta'} WAIT \ \delta - \delta'} \text{ } (\delta - Wait)$$

$$\frac{}{WAIT \ 0 \xRightarrow{\sqrt{\cdot}, t} E} \text{ (Wait)}$$

$$4. \frac{}{a \rightarrow P \xRightarrow{\delta} a \rightarrow P} \text{ } (\delta - Prefix)$$

$$\frac{}{a \rightarrow P \xRightarrow{a, t} P} \text{ (Prefix)}$$

$$\frac{}{([\sim_R y] \rightarrow P) : x[v] \xRightarrow{\langle x, x \sim_R y \rangle, t} P : x[v']} \text{ } (dm - Prefix),$$

where $v' = [x, x \sim_R y](v)$.

$$\frac{\forall i \in I. (x[v] \vdash_{\mathcal{A}, As} x \sim_{R_i} y_i), t = t'}{a @ (\bigwedge_{i \in I} \sim_{R_i} y_i, t') \rightarrow P : x[v] \xRightarrow{a, t} P : x[v]} \text{ } (s - Prefix)$$

$\delta - Prefix$ and $Prefix$ are similar to $\delta - Skip$ and $Skip$ respectively. We skip the explanations of them. In $s - Prefix$, action a is bounded with a space condition and time condition, it is allowed to occur only when all space and time conditions have been met.

$$5. \frac{C = true}{P <C> Q \xRightarrow{\tau, t} P} \text{ (Choice 1l)}$$

$$\frac{C = false}{P <C> Q \xRightarrow{\tau, t} Q} \text{ (Choice 1r)}$$

$$\frac{P \xRightarrow{\delta} P', Q \xRightarrow{\delta} Q'}{P \sqcap Q \xRightarrow{\delta} P' \sqcap Q'} \text{ } (\delta - Choice \ 2)$$

$$\frac{P : x[v] \xRightarrow{\theta, t} P' : x[v']}{(P \sqcap Q) : x[v] \xRightarrow{\theta, t} (P' \sqcap Q) : x[v']} \text{ } (\tau - Choice \ 2l)$$

$$\begin{array}{c}
\frac{Q : x[v] \xrightarrow{\theta, t} Q' : x[v']}{(P \sqcap Q) : x[v] \xrightarrow{\theta, t} (P \sqcap Q') : x[v']} (\tau - \text{Choice } 2r) \\
\frac{P \xrightarrow{e, t} P'}{P \sqcap Q \xrightarrow{e, t} P'} (\text{Choice } 2l) \\
\frac{Q \xrightarrow{e, t} Q'}{P \sqcap Q \xrightarrow{e, t} Q'} (\text{Choice } 2r) \\
\frac{P \xRightarrow{\delta} P', Q \xRightarrow{\delta} Q'}{P \sqcap Q \xRightarrow{\delta} P' \sqcap Q'} (\delta - \text{Choice } 3) \\
\frac{P : x[v] \xrightarrow{\omega, t} P' : x[v']}{(P \sqcap Q) : x[v] \xrightarrow{\omega, t} P' : x[v']} (\text{Choice } 3l) \\
\frac{Q : x[v] \xrightarrow{\omega, t} Q' : x[v']}{(P \sqcap Q) : x[v] \xrightarrow{\omega, t} Q' : x[v']} (\text{Choice } 3r)
\end{array}$$

Notice that in *Choice 1l, 1r*, the judgement of C is an internal action τ from the outside observers. From $\tau - \text{Choice } 2l, 2r$, a difference could be seen between $\sqcap$ and $\sqcap$. In $\sqcap$, the process leaves the choice unsolved until the environment participates in either P or Q . While in *Choice 3l, 3r*, the choice making is independent on the participation of the environment and any internal actions of P and Q can determine the choice.

$$\begin{array}{c}
6. \quad \frac{P \xRightarrow{\delta} P', \sqrt{\notin} \text{init}(P)}{P; Q \xRightarrow{\delta} P'; Q} (\delta - \text{Seq}) \\
\frac{P : x[v] \xrightarrow{\mu, t} P' : x[v']}{(P; Q) : x[v] \xrightarrow{\mu, t} (P'; Q) : x[v']} (\text{Seq}) \\
\frac{P \xrightarrow{\sqrt{\notin}, t} E}{P; Q \xrightarrow{\tau, t} Q} (\sqrt{\notin} - \text{Seq})
\end{array}$$

Since in $P; Q$, if $Q \neq E$, the *surd* of P is hidden as an internal action τ (as in $\sqrt{\notin} - \text{Seq}$ shown), which is forced to be triggered as soon as P is ready to engage in it. So in $\delta - \text{Seq}$ we add $\sqrt{\notin} \notin \text{init}P$.

$$\begin{array}{c}
7. \quad \frac{P \xRightarrow{\delta} P', Q \xRightarrow{\delta} Q'}{P_A \parallel_B Q \xRightarrow{\delta} P'_A \parallel_B Q'} (\delta - \text{Para}) \\
\frac{P : x[v] \xrightarrow{\omega, t} P' : x[v'], \omega \in A - B \vee \omega \in \Xi^\tau}{(P_A \parallel_B Q) : x[v] \xrightarrow{\omega, t} (P'_A \parallel_B Q) : x[v']} (\text{Para } l) \\
\frac{Q : x[v] \xrightarrow{\omega, t} Q' : x[v'], \omega \in B - A \vee \omega \in \Xi^\tau}{(P_A \parallel_B Q) : x[v] \xrightarrow{\omega, t} (P_A \parallel_B Q') : x[v']} (\text{Para } r) \\
\frac{P \xrightarrow{a, t} P', Q \xrightarrow{a, t} Q', a \in A \cap B}{P_A \parallel_B Q \xrightarrow{a, t} P'_A \parallel_B Q'} (\text{Para } lr)
\end{array}$$

Para lr indicates the synchronous communication between two processes. It could be the condition that one process wait

for another, allowed by rule $\delta - \text{Para}$.

$$\begin{array}{c}
8. \quad \frac{P \xRightarrow{\delta} P', Q \xRightarrow{\delta} Q'}{P \triangleleft Q \xRightarrow{\delta} P' \triangleleft Q'} (\delta - \text{Int}) \\
\frac{a_i \rightarrow P_i \xrightarrow{a_i, t} P_i}{P \triangleleft (\sqcap_{i \in I} a_i \rightarrow P_i) \xrightarrow{a_i, t} P_i} (\text{Int } 1) \\
\frac{P_1 < C > P_2 \xrightarrow{\tau, t} P_1}{P \triangleleft (P_1 < C > P_2) \xrightarrow{\tau, t} P_1} (\text{Int } 2) \\
\frac{P_1 < C > P_2 \xrightarrow{\tau, t} P_2}{P \triangleleft (P_1 < C > P_2) \xrightarrow{\tau, t} P; P_2} (\text{Int } 3)
\end{array}$$

In *Int 3*, if $C = \text{false}$, then P will not be interrupted. P_2 will proceed after the finish of P .

$$\begin{array}{c}
9. \quad \frac{(\alpha \rightarrow P) : x[v] \xrightarrow{\alpha, t} P : x[v'], \alpha \in A}{((\alpha \rightarrow P)/A) : x[v] \xrightarrow{\tau, t} (P/A) : x[v']} (\delta - \text{Abs}) \\
\frac{(\alpha \rightarrow P) : x[v] \xrightarrow{\alpha, t} P : x[v'], \alpha \notin A}{((\alpha \rightarrow P)/A) : x[v] \xrightarrow{\alpha, t} (P/A) : x[v']} (\text{Abs}) \\
10. \quad \frac{}{\mu X. P(X) \xrightarrow{\tau} P(\mu X. P(X)/X)} (\text{Rec}) \\
11. \quad \frac{P \xRightarrow{\delta} P'}{g(P) \xRightarrow{\delta} g(P')} (\delta - \text{Ren}) \\
\frac{P : x[v] \xrightarrow{\omega, t} P' : x[v']}{g(P) : x[v] \xrightarrow{g(\omega), t} g(P') : x[v']} (\delta - \text{Ren}),
\end{array}$$

where $g \in \{f, f^{-1}\}$.

$$\begin{array}{c}
12. \quad \frac{M_1 \xRightarrow{\delta} M'_1, \dots, M_n \xRightarrow{\delta} M'_n}{|(M_1, \dots, M_n) \xRightarrow{\delta} |(M'_1, \dots, M'_n)} (\delta - \text{agpara}) \\
\frac{P_i : x_i[v] \xrightarrow{\langle x_i, x_i \sim R y_i \rangle, t} P'_i : x_i[v']}{|(P_1 : x_1[v], \dots, P_i : x_i[v], \dots, P_n : x_n[v]) \xrightarrow{\langle x_i, x_i \sim R y_i \rangle, t} |(P_1 : x_1[v'], \dots, P'_i : x_i[v'], \dots, P_n : x_n[v'])}, \\
\text{where } 1 \leq i \leq n.
\end{array}$$

where $1 \leq i \leq n$.

$$\begin{array}{c}
\frac{M_i \xrightarrow{\omega, t} M'_i, \omega \notin A \cup \Xi}{|(M_1, \dots, M_i, \dots, M_n) \xrightarrow{\omega, t} |(M_1, \dots, M'_i, \dots, M_n),} \\
\text{where } 1 \leq i \leq n.
\end{array}$$

where $1 \leq i \leq n$.

$$\begin{array}{c}
M_i \xrightarrow{a!v, t} M'_i, \\
M_{j_1} \xrightarrow{a?v_1, t} M'_{j_1}, \dots, M_{j_l} \xrightarrow{a?v_l, t} M'_{j_l}, \\
M_{j_{l+1}} \xrightarrow{a?v_{l+1}, t} M'_{j_{l+1}}, \dots, M_{j_m} \xrightarrow{a?v_m, t} M'_{j_m}, a \in A
\end{array}$$

$$\begin{aligned}
S_{ex1} &\xrightarrow[\emptyset]{\tau^2,0,*} |((WAIT\ 11; [\mathbf{LE}^{-1}25] \rightarrow WAIT\ 13; [\mathbf{LT}10] \rightarrow \mu X.F_1(X)) : T[\emptyset], \\
&(WAIT\ 0.1; (send@(\mathbf{LT}5) \rightarrow \mu X.F_2(X)) \square (idle \rightarrow \mu X.F_2(X))) : T[\emptyset]) \\
&\xrightarrow[\emptyset]{(idle \cdot \tau)^{110},11,*} |((WAIT\ 0; [\mathbf{LE}^{-1}25] \rightarrow WAIT\ 13; [\mathbf{LT}10] \rightarrow \mu X.F_1(X)) : T[\emptyset], \\
&(WAIT\ 0; (send@(\mathbf{LT}5) \rightarrow \mu X.F_2(X)) \square (idle \rightarrow \mu X.F_2(X))) : T[\emptyset]) \\
&\xrightarrow[\emptyset]{\tau^2,11,*} |([\mathbf{LE}^{-1}25] \rightarrow WAIT\ 13; [\mathbf{LT}10] \rightarrow \mu X.F_1(X)) : T[\emptyset], \\
&((send@(\mathbf{LT}5) \rightarrow \mu X.F_2(X)) \square (idle \rightarrow \mu X.F_2(X))) : T[\emptyset]) \\
&\xrightarrow[\emptyset]{\langle T, T\mathbf{LE}^{-1}25 \rangle \cdot idle, 11,*} |((WAIT\ 13; [\mathbf{LT}10] \rightarrow \mu X.F_1(X)) : T[\{T\mathbf{LE}^{-1}25\}], (\mu X.F_2(X)) : T[\{T\mathbf{LE}^{-1}25\}]) \\
&\xrightarrow[\emptyset]{\tau \cdot (idle \cdot \tau)^{130}, 24,*} |((WAIT\ 0; [\mathbf{LT}10] \rightarrow \mu X.F_1(X)) : T[\{T\mathbf{LE}^{-1}25\}], \\
&(WAIT\ 0; (send@(\mathbf{LT}5) \rightarrow \mu X.F_2(X)) \square (idle \rightarrow \mu X.F_2(X))) : \langle T, \{T\mathbf{LE}^{-1}25\} \rangle) \\
&\xrightarrow[\emptyset]{\tau^2, 24,*} |([\mathbf{LT}10] \rightarrow \mu X.F_1(X)) : T[\{T\mathbf{LE}^{-1}25\}], \\
&((send@(\mathbf{LT}5) \rightarrow \mu X.F_2(X)) \square (idle \rightarrow \mu X.F_2(X))) : T[\{T\mathbf{LE}^{-1}25\}]) \\
&\xrightarrow[\emptyset]{\langle T, T\mathbf{LT}10 \rangle, 24,*} |(\mu X.F_1(X) : T[\{T\mathbf{LT}10\}], ((send@(\mathbf{LT}5) \rightarrow \mu X.F_2(X)) \square (idle \rightarrow \mu X.F_2(X))) : T[\{T\mathbf{LT}10\}]) \\
&\xrightarrow[\emptyset]{send, 24,*} |(\mu X.F_1(X) : T[\{T\mathbf{LT}10\}], \mu X.F_2(X) : T[\{T\mathbf{LT}10\}]) \\
&\xrightarrow[\emptyset]{\tau^2, 24,*} \dots
\end{aligned}$$

FIGURE 4. One possible transition of S_{ex1} .

$$\begin{aligned}
&\frac{}{A \vdash (\dots, M'_{j_1}, \dots, M'_{j_l}, M'_i, M'_{j_{l+1}}, \dots, M'_{j_m}, \dots) \text{ (agpara } b)} \\
&\xrightarrow[A]{a.v.t} |(\dots, M'_{j_1}, \dots, M'_{j_l}, M'_i, M'_{j_{l+1}}, \dots, M'_{j_m}, \dots),
\end{aligned}$$

where $1 \leq i \leq n$, $1 \leq m \leq n-1$, $1 \leq j_1 \leq \dots \leq j_l < i < j_{l+1} \leq \dots \leq j_m \leq n$.

dm – *agpara* rule shows how the dynamic mapping changes the region relations between agents. The broadcasting rule *agpara b* shows that when one agent sends through a channel a , all other agents receive through the same channel simultaneously.

VI. EXAMPLES

In this section, we will show QCSP can model the spatio-temporal behavior of the 3 examples introduced in section I by selecting suitable qualitative calculus to capture the topological structure of spaces.

A. EXAMPLE 1: TEMPERATURE SENSOR

Consider a temperature sensor in a wireless sensor network. Assuming that the temperature changes itself periodically to over 25 degree in 11 hours then lower than 5 degree in the rest 13 hours in a day. The sensor keeps sensing the temperature every 0.1 hour and only sends alert when the temperature is lower than 10 degree or keeps idling otherwise.

We adopt the point algebra (section III-B.1) as the qualitative calculus. However, since the universal region set U

now becomes $\mathbb{R}$, we have to add an assumption in point algebra:

$$p : \forall x, y \in \mathbb{R}. (x < y \leftrightarrow x\mathbf{LT}y),$$

which says operator $\mathbf{LT}$ exactly corresponds to $<$ in $\mathbb{R}$. Thus we set $As'_{PA} = \{p\}$ and consider $\langle \mathcal{A}_{PA}, As'_{PA} \rangle$ as our qualitative calculus.

$$S_{ex1} = |(\text{Tem}, \text{Sen}),$$

where

$$\begin{aligned}
\text{Tem} &= \mu X.F_1(X) : T[\emptyset] = \mu X.(WAIT\ 11; [\mathbf{LE}^{-1}25] \\
&\rightarrow WAIT\ 13; [\mathbf{LT}10] \rightarrow X) : T[\emptyset], \\
\text{Sen} &= \mu X.F_2(X) : T[\emptyset] = \mu X.(WAIT\ 0.1; \\
&(send@(\mathbf{LT}5) \rightarrow X) \square (idle \rightarrow X)) : T[\emptyset]
\end{aligned}$$

Temperature values in $\mathbb{R}$ are all assumed to be space regions under $\langle \mathcal{A}_{PA}, \emptyset \rangle$. We set an abstract region T as the current temperature value for both Tem and Sen . At the beginning, there is no any region relations in the system, so we set the constraints network to $\emptyset$. Agent Tem takes 11 hours to move in region T to some ‘point’ more than 25, then takes 13 hours to move to less than 10. Agent Sen waits for every 0.1 hours, either sending data if T is less than 5 (i.e., has the relation $T\mathbf{LT}5$), or keeping idling.

With operational semantics, we can give one possible transition of S_{ex1} shown in Fig. 4.

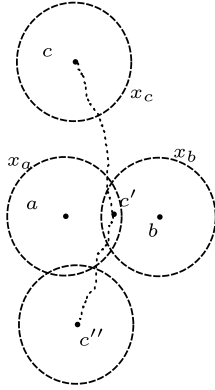


FIGURE 5. Three nodes in mobile ad hoc network.

B. EXAMPLE 2: MOBILE AD HOC NETWORK

In a mobile ad hoc network, each node has a transmission range, only in which other nodes can receive messages from it. The communication between nodes is symmetric, i.e., if node a is in the transmission range of node b , node b must be in the transmission range of node a too. Each node can move freely at any time from one location to others. Fig. 5 shows three nodes a, b, c in a network. x_a, x_b, x_c are their transmission ranges — the space regions. The transmission ranges of node a and b intersect with each other. Node a (b) is broadcasting periodically every 1 seconds, while node c starts at a place out of transmission ranges of a and b and moves across the intersecting part of a and b .

For this example one problem that needs to be considered is to define an appropriate qualitative calculus to express the ‘communication relations’ between nodes. Since in QCSP, one agent can only belong to one region, classical topological space calculus like RCC5/8 algebra [36] cannot meet our demands here because we cannot distinguish the difference between the node itself (as a point) and its transmission range by only showing the topological region relations between transmission ranges. For example, $x_a \mathbf{O} x_b$ in RCC8 means ‘ x_a overlaps x_b ’, which does not indicate node a can communicate with b (see the locations of node a and b in Fig. 5).

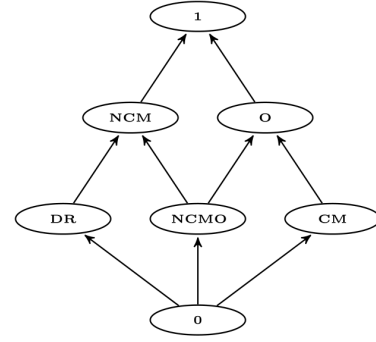
For this reason we give a special qualitative calculus for reasoning about the symmetric communication relation between nodes.

Definition 6.1 (Symmetric Communication Algebra): A qualitative calculus called ‘symmetric communication algebra’ $\langle \mathcal{A}_{SCA}, As_{SCA} \rangle$ is given as follows.

Let U be a space region, $x, y \in U$ be any space regions. Define $x \mathbf{CM} y$ as ‘ x can communicate with y ’. $\mathbf{CM}$ is reflexive and symmetric, so we have propositions p_1, p_2 as two assumptions of this calculus as follows:

$$\begin{aligned} p_1 &: \forall x \in U. (x \mathbf{CM} x); \\ p_2 &: \forall xy. (x \mathbf{CM} y \rightarrow y \mathbf{CM} x). \end{aligned}$$

Let $As_{SCA} = \{p_1, p_2\}$.

FIGURE 6. The lattice structure of $\mathcal{A}_{SCA}$.

Other relations in $\mathcal{A}_{SCA}$ are given as:

$$\begin{aligned} x \mathbf{NCM} y &\triangleq \neg(x \mathbf{CM} y); \\ x \mathbf{O} y &\triangleq \exists z. (x \mathbf{CM} z \wedge z \mathbf{CM} y); \\ x \mathbf{NCMO} y &\triangleq x \mathbf{O} y \wedge \neg(x \mathbf{CM} y); \\ x \mathbf{DR} y &\triangleq \neg(x \mathbf{O} y). \end{aligned}$$

The basic relations of $\mathcal{A}_{SCA}$ is $\mathcal{B}_{SCA} = \{\mathbf{DR}, \mathbf{NCMO}, \mathbf{CM}\}$.

$x \mathbf{NCM} y$ means x cannot communicate with y . Not like in RCC8, $x \mathbf{O} y$ here is defined as the composition of $\mathbf{CM}$ and $\mathbf{NCMO}$ (i.e., $\mathbf{O} = \mathbf{CM} \cup \mathbf{NCMO}$). It means x overlaps y . $x \mathbf{NCMO} y$ means that x overlaps y but x cannot communicate with y . $x \mathbf{DR} y$ means that x is distant from y .

The lattice structure of $\mathcal{A}_{SCA}$ is shown in Fig. 6. With $\langle \mathcal{A}_{SCA}, As_{SCA} \rangle$ we can give the formal model of the mobile ad hoc network system as follows:

$$S_{ex2} = \mid_{\{c_a, c_b\}} (N_a, N_b, N_c)$$

where

$$\begin{aligned} N_a &= \mu X. (\text{WAIT } 1; c_a!v_a \rightarrow X) \setminus \Xi : x_a[v], \\ N_b &= \mu X. (\text{WAIT } 1; c_b!v_b \rightarrow X) \setminus \Xi : x_b[v], \\ N_c &= (P_1 \parallel P_2) \setminus \Xi : x_c[v]. \end{aligned}$$

$$v = \{x_a \mathbf{O} x_b, x_a \mathbf{DR} x_c, x_b \mathbf{DR} x_c\},$$

$$\begin{aligned} P_1 &= \mu X. (\text{WAIT } 1; (c_a @ (\mathbf{CM} x_a) ? v \\ &\quad \square c_b @ (\mathbf{CM} x_b) ? v \square \text{SKIP}); X), \end{aligned}$$

$$\begin{aligned} P_2(\delta_1, \delta_2) &= \text{WAIT } \delta_1; [\mathbf{CM} x_a] \rightarrow [\mathbf{CM} x_b] \rightarrow \text{WAIT } \delta_2; \\ &\quad [\mathbf{NCMO} x_a] \rightarrow [\mathbf{DR} x_b] \rightarrow \text{SKIP}. \end{aligned}$$

Node a, b, c are confined in their space regions x_a, x_b, x_c respectively. Nodes are independent so their movements are hidden from others. The initial value of v shows that the related locations of three nodes at the beginning. The behavior of node a and b is similar. Node a broadcasts message v_a through channel c_a every 1 second. The behavior of node c constitutes two parts: P_1 and P_2 . P_1 is a loop, it receives message from node a or b when it is in the transmission range of them respectively. P_2 is a parameterized process with two variables δ_1, δ_2 , it indicates the movement of node c .

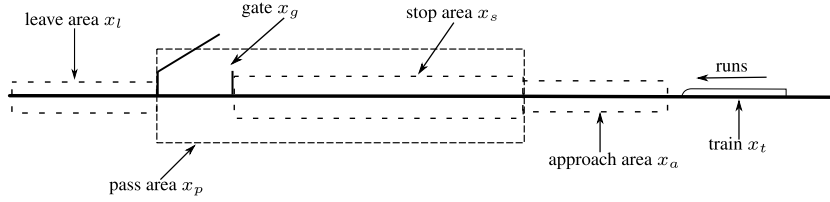


FIGURE 7. Intelligent railroad crossing control system.

Node c first moves in the range of a and b for δ_1 seconds, then moves out of ranges of a and b (but still overlaps a) in δ_2 seconds (Fig. 5 shows the trajectory of node c).

C. EXAMPLE 3: INTELLIGENT RAILROAD CROSSING CONTROL SYSTEM

Intelligent Railroad Crossing Control System is an intelligent transportation system. Fig. 7 shows a plane graph of the scenario of this system. A train runs on a track, it approaches a crossing where an intelligent gate agent is built to keep safe. A road lies across the gate in north-south direction. When the gate is open, it allows vehicles on the road to pass through. When it is closed, it stops vehicles and allows the train on the track to pass through. The system describes a simplified scenario of interactions between the gate and the train in reality.

There are several critical areas — space regions on the track, the train must behave accordingly based on these areas in order to keep safe. The ‘approach area’ (denoted by x_a) is the area where a train can send and receive signals from the gate. It indicates that the train must complete its communicating with the gate in this area, otherwise it would be very dangerous since the train might not behave correctly in time after leaving this area and entering the ‘stop area’. ‘Stop area’ x_s is the area for a train to stop if it gets the gate’s message that ‘it cannot pass through’. It could happen if the gate cannot be closed because there are still cars passing through. The same, the train must be stopped in this area or a collision might happen. The area that embodies the stop area and the gate area x_g is the ‘pass area’ x_p . No matter the gate stops the train or not, the train must pass through the crossing in this area. There is a ‘leave area’ x_l indicating that a train has already passed through. It is useful since the gate will know when a train has already passed and can be reopened for vehicles to pass. The train and gate are confined in a region of themselves by x_l and x_g respectively.

To reason about the topological space regions we adopt interval algebra (section III-B.2). However, we need to give several assumptions since all intervals in this example have fixed sizes and locations. The length of train is usually longer than the gate. We assume that all areas (except the gate) are assumed to be longer than the length of train. Moreover, the relative location relations among the gate and critical areas are fixed.

The additional assumptions in interval algebra is given below:

$$\begin{aligned} p_1 &: \neg(x_t \{S, DU, E\} x_g); \\ p_2 &: \forall x \in \{x_s, x_l, x_p, x_a\}. (\neg(x \{S, DU, E\} x_t)); \\ p_3 &: x_l MT x_g \wedge x_g MT x_s \wedge x_s MT x_a; \\ p_4 &: x_l MT x_p \wedge x_p MT x_a; \\ p_5 &: x_g S x_p \wedge x_s E x_p. \end{aligned}$$

p_1 specifies that the train’s interval can never be a part of the gate’s interval. p_2 indicates all critical areas except for x_g are longer than a train. p_3, p_4, p_5 give the location relations of the gate and critical areas.

We obtain a new assumptions set $As'_{IA} = As_{IA} \cup \{p_1, p_2, p_3, p_4, p_5\}$ and adopt $\langle \mathcal{A}_{IA}, As'_{IA} \rangle$ as our qualitative calculus.

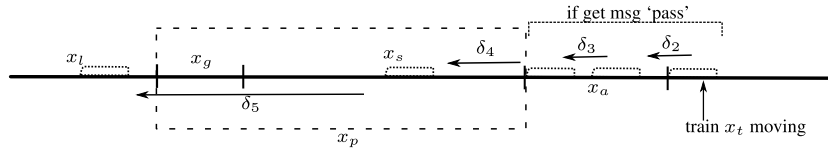
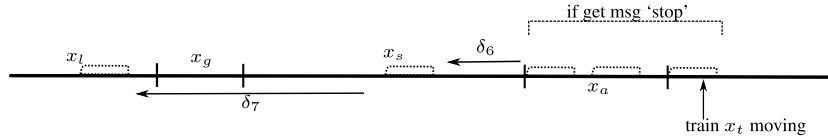
$$S_{ex3} = \mid_{\{a\}} (T, G)$$

where

$$\begin{aligned} T &= ([x_a MT] \rightarrow \left(\begin{array}{l} P_1 \parallel \\ a! \text{'arrive'} \rightarrow a@(\{MT^{-1}, S, \\ DU, E\} x_a)?v \rightarrow SKIP \end{array} \right) \\ &\quad \triangleright P_2 <v = \text{'stop'}> SKIP) \setminus \Xi : x_t[v], \\ G &= \mu X. (WAIT \ 1; a?v \rightarrow Q_1 <v = \text{'arrive'}> \\ &\quad (Q_2 <v = \text{'leave'}> X)) \setminus A : x_g[v], \\ &\quad \text{where } A = \{open, op, cl, ccl\}. \\ P_1 &= WAIT \ \delta_2; [\{E, DU\} x_a] \rightarrow WAIT \ \delta_3; [x_p MT] \\ &\quad \rightarrow WAIT \ \delta_4; [\{E, DU, S\} x_p] \rightarrow WAIT \ \delta_5; \\ &\quad [\{E, DU\} x_l] \rightarrow SKIP, \\ P_2 &= WAIT \ \delta_6; [\{E, DU, S\} x_s] \rightarrow a? \text{'pass'} \\ &\quad \rightarrow WAIT \ \delta_7; [\{E, DU\} x_l] \rightarrow a! \text{'leave'} \rightarrow SKIP, \\ Q_1 &= \left(\begin{array}{l} op \rightarrow a! \text{'stop'} \rightarrow \mu Y. (WAIT \ \theta_1; \\ (ccl \rightarrow WAIT \ \theta_2; Q_3) \sqcap Y) \\ \sqcap \\ cl \rightarrow Q_3 \end{array} \right), \\ Q_2 &= WAIT \ \theta_3; open \rightarrow X, \\ Q_3 &= a! \text{'pass'} \rightarrow X. \end{aligned}$$

As independent agents the train and gate’s behaviors are hidden from each other. They only interact with each other by communications.

The train at some time point arrives in area x_a , at which it sends message ‘arrive’ through channel a to the gate while

FIGURE 8. Train's behavior in P_1 .FIGURE 9. Train's behavior in P_2 .

keeping moving following a movement pattern P_1 . In P_1 , the train keep approaching to the passing area x_p with 'some speed' which we do not care. But we know that it would take δ_2 seconds to move into the approach area x_a and forward about to enter x_p for δ_3 seconds. After that it keeps moving and totally enters x_p in δ_4 seconds, and then finally passes the gate in δ_5 seconds. Fig 8 gives a 'motion locus' of the train following the pattern P_1 .

While the train is moving, it is waiting for the signals from the gate. Only when it gets signals in the area of x_a it is safe (because train might be too fast and cannot stop timely before it reaches the gate). If it gets the 'stop' message from the gate in 'safe area', its movement behavior P_1 is interrupted by an emergency process P_2 . P_2 would break down the train in the stop area in δ_6 seconds (no matter the exact current location of train). The train stops in area x_s and waits for message 'pass' from the gate. After that it continues to pass the gate and leaves the crossing. Fig. 9 shows the condition if train takes P_2 .

The gate is located at the crossing, every 1 second it receives signals from the train. If getting message 'arrive', the gate executes the subprocess Q_1 , else if getting message 'leave', the gate executes the subprocess Q_2 . Q_1 describes the gate's behavior of judging if a train can safely pass. Action op and cl is a judgement of whether gate is closed or open. Whether the gate is open or closed depends on current traffic condition on the road, thus it is modelled as a non-deterministic choice in QCSP. If the gate is open, the gate proceeds Q_3 , sending message 'pass' to inform the train to safely pass. If the gate is open, it sends message 'stop' to stop the train and goes into a loop $\mu Y. (...Y...)$. In the loop the gate judges the traffic condition every θ_1 time, if there is no vehicles on the road so that the gate can be closed (indicated by action ccl), it would take θ_2 time to close the gate and then go on Q_3 . Q_2 describes the behavior that the gate would take θ_3 to open after one train has successfully passed the crossing.

VII. CONCLUSION AND FUTURE WORK

In this paper, we propose a process algebra called QCSP for formal modeling systems that are highly sensitive to space

and time. We propose the dynamic mapping to represent the mobility of space regions. We give a formal definition of QCSP and its operational semantics in the sense of labelled transition system. At last, we illustrate the application of QCSP in modeling the spatio-temporal behavior by three examples.

In QCSP, spatio-temporal constraints are captured and reasoned by the qualitative relations between the space regions of agents. Qualitative calculus manages to capture the topological structure of spaces in a finite set of relations. A set of relations between space regions forms a constraints network, which together with assumptions forms the basis of reasoning spaces. The dynamic movement of agents is represented through the dynamic change of constraints network — the dynamic mapping, which maps a consistent network to the maximum consistent network of the network formed by adding a new relation into the original one. In section IV, we proved the existence and the uniqueness of the maximum consistent network. The dynamic mapping is embedded in QCSP as a primitive to express the moving capabilities.

Comparing with π -calculus [17], AC [18] and other calculus based on them [20]–[23], [25], [27], [28], QCSP is equipped with a more general ability to express qualitative relations. In fact, the space relations in π -calculus and AC are just a special case in QCSP by selecting a 'suitable' boolean algebra (illustrated in the caption of Table 1). Qualitative calculus in QCSP offers to capture different kinds of topological spaces with different boolean algebras. As shown in section VI, we use three different algebras for modeling three different systems, whose spaces' topological structures vary from one to another.

The denotational semantics and verification aspects should be fully investigated after building the theory of QCSP. We will focus on the verification aspects of QCSP and compare it with previous calculus like Timed CSP or π -CCS for further study.

REFERENCES

- [1] E. A. Lee, "Cyber physical systems: Design challenges," in *Proc. IEEE 11th Int. Symp. Object Compon.-Oriented Real-Time Distrib. Comput. (ISORC)*, May 2008, pp. 363–369.

- [2] B. N. Kholodenko, "Cell-signalling dynamics in time and space," *Nature Rev. Molecular Cell Biol.*, vol. 7, no. 3, pp. 165–176, 2006.
- [3] I. F. Akyildiz and I. H. Kasimoglu, "Wireless sensor and actor networks: Research challenges," *Ad Hoc Netw.*, vol. 2, no. 4, pp. 351–367, Oct. 2004.
- [4] S. Corson and J. Macker, "Mobile ad hoc networking (MANET): Routing protocol performance issues and evaluation considerations," Tech. Rep., 1999.
- [5] Y. Chen, "STeC: A location-triggered specification language for real-time systems," in *Proc. IEEE 15th Int. Symp. Object/Compon./Service-Oriented Real-Time Distrib. Comput. Workshops (ISORC)*, Apr. 2012, pp. 1–6.
- [6] G. D. Abowd, A. K. De, P. J. Brown, N. Davies, M. Smith, and P. Steggles, "Towards a better understanding of context and context-awareness," in *Proc. Int. Symp. Handheld Ubiquitous Comput. (HUC)*, 1999, pp. 304–307.
- [7] M. Aiello, I. Pratt-Hartmann, and J. Van Benthem, "What is spatial logic?" in *Handbook Spatial Logics*. Springer, 2007, pp. 1–11.
- [8] G. Ligozat and J. Renz, "What is a qualitative calculus? A general framework," in *Proc. Pacific Rim Int. Conf. Artif. Intell. (PRICAI)*, 2004, pp. 53–64.
- [9] C. Landsiedel, V. Rieser, M. Walter, and D. Wollherr, "A review of spatial reasoning and interaction for real-world robotics," *Adv. Robot.*, vol. 31, pp. 222–242, Jan. 2017.
- [10] E. Irwansyah, *Geographic Information System*. Hoboken, NJ, USA: Wiley, 2017.
- [11] G. M. Reed and A. W. Roscoe, "A timed model for communicating sequential processes," *Theor. Comput. Sci.*, vol. 58, pp. 249–261, 1988.
- [12] C. A. R. Hoare, "Communicating sequential processes," *Commun. ACM*, vol. 21, no. 8, pp. 666–677, 1978.
- [13] R. Milner, *A Calculus of Communicating Systems*. New York, NY, USA: Springer-Verlag, 1982.
- [14] J. A. Bergstra and J. W. Klop, "Algebra of communicating processes with abstraction," *Theor. Comput. Sci.*, vol. 37, pp. 77–121, 1985.
- [15] G. Berry and G. Boudol, "The chemical abstract machine," *Theor. Comput. Sci.*, vol. 96, no. 1, pp. 217–248, 1992.
- [16] J. C. M. Baeten and J. A. Bergstra, "Real space process algebra," *Formal Aspects Comput.*, vol. 5, no. 6, pp. 481–529, 1993.
- [17] R. Milner, J. Parrow, and D. Walker, "A calculus of mobile processes, Parts I and II," *Inf. Comput.*, vol. 100, no. 1, pp. 1–77, 1992.
- [18] L. Cardelli and A. D. Gordon, "Mobile ambients," *Electron. Notes Theor. Comput. Sci.*, vol. 10, pp. 198–201, 1997.
- [19] Y. Zhang, F. Mallet, and Y. Chen, "A verification framework for spatio-temporal consistency language with CCSL as a specification language," in *Frontiers of Computer Science*. 2018. [Online]. Available: http://journal.hep.com.cn/fcs/EN/abstract/article_21720.shtml
- [20] M. John, R. Ewald, and A. M. Uhrmacher, "A spatial extension to the π calculus," *Electron. Notes Theor. Comput. Sci.*, vol. 194, no. 3, pp. 133–148, 2008.
- [21] A. Singh, C. R. Ramakrishnan, and S. A. Smolka, "A process calculus for mobile ad hoc networks," *Sci. Comput. Program.*, vol. 75, no. 6, pp. 440–469, 2010.
- [22] N. Gul, "A calculus of mobility and communication for ubiquitous computing," *Electron. Proc. Theor. Comput. Sci.*, vol. 188, p. 8, 2015.
- [23] R. Lanotte and M. Merro, "A semantic theory of the Internet of Things," in *Proc. Int. Conf. Coordination Models Lang. (COORDINATION)*, 2016, pp. 157–174.
- [24] C. Brooks *et al.*, "A component architecture for the Internet of Things," *Proc. IEEE*, vol. 106, no. 9, pp. 1527–1542, Sep. 2018.
- [25] Y. A. Alrahman, R. De Nicola, and M. Loreti, "A behavioural theory for interactions in collective-adaptive systems," *CoRR*, 2017.
- [26] A. Ferscha, "Collective adaptive systems," in *Proc. ACM Int. Joint Conf. Pervasive Ubiquitous Comput., Int. Symp. Wearable Comput. (UbiComp/ISWC)*, 2015, pp. 893–895.
- [27] F. Sieue, H. Zedan, and A. Cau, "The calculus of context-aware ambients," *J. Comput. Syst. Sci.*, vol. 77, no. 4, pp. 597–620, 2011.
- [28] D. P. Pasqualin, J. K. Vizzotto, and E. K. Piveta, "Typed context awareness ambient calculus for pervasive applications," *Formal Aspects Comput.*, vol. 27, no. 5, pp. 885–916, 2015.
- [29] P. C. Attie and N. A. Lynch, "Dynamic input/output automata: A formal and compositional model for dynamic systems," *Inf. Comput.*, vol. 249, pp. 28–75, Aug. 2016.
- [30] A. Gassara, I. B. Rodriguez, and M. Jmaiel, "Towards a multi-scale modeling for architectural deployment based on bigraphs," in *Proc. Eur. Conf. Softw. Archit. (ECSA)*, 2013, pp. 122–129.
- [31] S. Han and H. Y. Youn, "Petri net-based context modeling for context-aware systems," *Artif. Intell. Rev.*, vol. 37, no. 1, pp. 43–67, 2012.
- [32] H. Hu, M. Zhou, and Y. Liu, "Supervisor synthesis and performance improvement for automated manufacturing systems by using Petri nets," in *Proc. IEEE Int. Conf. Autom. Sci. Eng. (CASE)*, Aug. 2013, pp. 1139–1144.
- [33] H. Hu, Y. Liu, and M. Zhou, "Maximally permissive distributed control of large scale automated manufacturing systems modeled with petri nets," *IEEE Trans. Control Syst. Technol.*, vol. 23, no. 5, pp. 2026–2034, Sep. 2015.
- [34] J. F. Allen, "Maintaining knowledge about temporal intervals," *ACM Commun.*, vol. 26, no. 11, pp. 832–843, Nov. 1983.
- [35] M. Vilain and H. Kautz, "Constraint propagation algorithms for temporal reasoning," in *Proc. Nat. Conf. Artif. Intell. (AAAI)*, 1986, pp. 377–382.
- [36] D. A. Randell, Z. Cui, and A. G. Cohn, "A spatial logic based on regions and connection," in *Proc. Int. Conf. Principles Knowl. Represent. Reasoning (KR)*, 1992, pp. 165–176.
- [37] A. W. Roscoe, *Theory and Practice of Concurrency*. Upper Saddle River, NJ, USA: Prentice-Hall, 1997.
- [38] S. Schneider, *Concurrent and Real-Time Systems: The CSP Approach*, 1st ed. Hoboken, NJ, USA: Wiley, 1999.
- [39] S. Schneider, "An operational semantics for timed CSP," *Inf. Comput.*, vol. 116, no. 2, pp. 193–213, 1995.



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